

Subject Notebook Template

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1 Usage

1.1 Highlighting

Text can be highlighted like this.

1.2 Theorems and Proofs

Theorem 1.1. *For any two real numbers a and b , the following inequality holds:*

$$(a + b)^2 \geq 4ab.$$

Proof. Expanding $(a - b)^2 \geq 0$, we get:

$$a^2 - 2ab + b^2 \geq 0 \implies a^2 + b^2 \geq 2ab \implies (a + b)^2 \geq 4ab.$$

□

1.3 Example

Example 1.2. Consider the function $f(x) = x^2$. For $x = 2$, we have $f(2) = 4$.